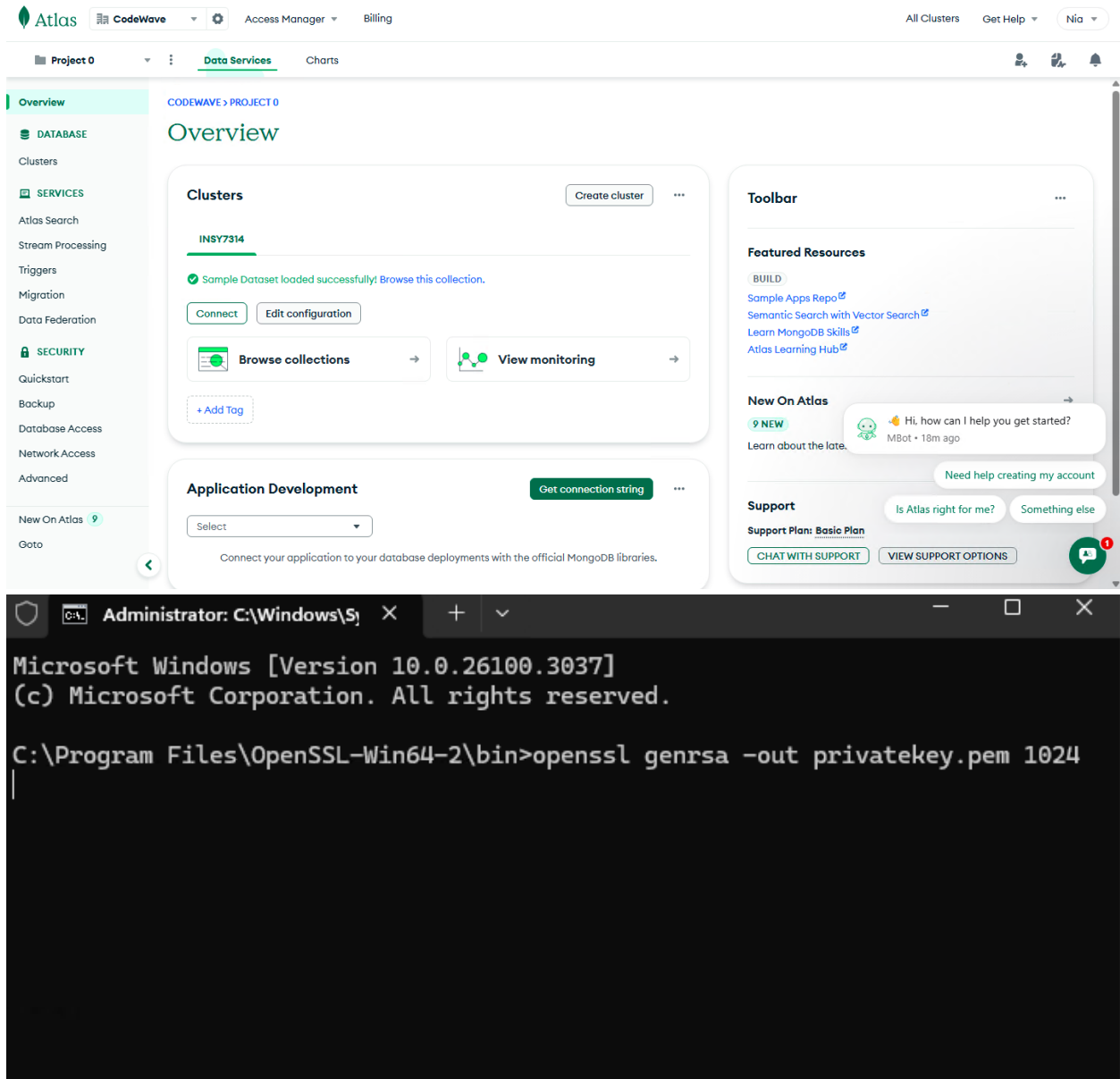


ICE TASK 1

ST10396650

INSY7314

SCREENSHOTS





Administrator: C:\Windows\Sy



Microsoft Windows [Version 10.0.26100.3037]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\OpenSSL-Win64-2\bin>openssl genrsa -out privatekey.pem 1024

C:\Program Files\OpenSSL-Win64-2\bin>dir privatekey.pem
Volume in drive C is Windows
Volume Serial Number is DCB9-6A0D

Directory of C:\Program Files\OpenSSL-Win64-2\bin

09/24/2025	01:25 PM	928	privatekey.pem
	1 File(s)	928	bytes
	0 Dir(s)	94,840,438,784	bytes free

C:\Program Files\OpenSSL-Win64-2\bin>

```
Administrator: C:\Windows\Sy x + v
Microsoft Windows [Version 10.0.26100.3037]
(c) Microsoft Corporation. All rights reserved.

C:\Program Files\OpenSSL-Win64-2\bin>openssl genrsa -out privatekey.pem 1024

C:\Program Files\OpenSSL-Win64-2\bin>dir privatekey.pem
Volume in drive C is Windows
Volume Serial Number is DCB9-6A0D

Directory of C:\Program Files\OpenSSL-Win64-2\bin

09/24/2025 01:25 PM          928 privatekey.pem
                1 File(s)          928 bytes
                0 Dir(s)  94,840,438,784 bytes free

C:\Program Files\OpenSSL-Win64-2\bin>openssl req -new -key privatekey.pem -o
ut certrequest.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:ZA
State or Province Name (full name) [Some-State]:GP
Locality Name (eg, city) []:PTA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:APDS
Organizational Unit Name (eg, section) []:STUDENT
Common Name (e.g. server FQDN or YOUR name) []:NIA
Email Address []:st10396650@vccconnect.edu.za

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:friedgreentomatoes
An optional company name []:
```

```
C:\Program Files\OpenSSL-Win64-2\bin>dir privatekey.pem
Volume in drive C is Windows
Volume Serial Number is DCB9-6A0D
```

```
Directory of C:\Program Files\OpenSSL-Win64-2\bin
```

```
09/24/2025  01:25 PM                928 privatekey.pem
               1 File(s)                928 bytes
               0 Dir(s)  94,840,438,784 bytes free
```

```
C:\Program Files\OpenSSL-Win64-2\bin>openssl req -new -key privatekey.pem -o
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```

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There are quite a few fields but you can leave some blank
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If you enter '.', the field will be left blank.

```
-----
Country Name (2 letter code) [AU]:ZA
State or Province Name (full name) [Some-State]:GP
Locality Name (eg, city) []:PTA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:APDS
Organizational Unit Name (eg, section) []:STUDENT
Common Name (e.g. server FQDN or YOUR name) []:NIA
Email Address []:st10396650@vcconnect.edu.za
```

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:friedgreentomatoes
An optional company name []:

```
C:\Program Files\OpenSSL-Win64-2\bin>openssl x509 -req -in certrequest.csr -
signkey privatekey.pem -out certificate.pem
Certificate request self-signature ok
subject=C=ZA, ST=GP, L=PTA, O=APDS, OU=STUDENT, CN=NIA, emailAddress=st10396
650@vcconnect.edu.za
```

```
C:\Program Files\OpenSSL-Win64-2\bin>
```

```
C:\Program Files\OpenSSL-Win64-2\bin>openssl req -new -key privatekey.pem -o  
ut certrequest.csr
```

You are about to be asked to enter information that will be incorporated
into your certificate request.

What you are about to enter is what is called a Distinguished Name or a DN.

There are quite a few fields but you can leave some blank

For some fields there will be a default value,

If you enter '.', the field will be left blank.

Country Name (2 letter code) [AU]:ZA

State or Province Name (full name) [Some-State]:GP

Locality Name (eg, city) []:PTA

Organization Name (eg, company) [Internet Widgits Pty Ltd]:APDS

Organizational Unit Name (eg, section) []:STUDENT

Common Name (e.g. server FQDN or YOUR name) []:NIA

Email Address []:st10396650@vcconnect.edu.za

Please enter the following 'extra' attributes
to be sent with your certificate request

A challenge password []:friedgreentomatoes

An optional company name []:

```
C:\Program Files\OpenSSL-Win64-2\bin>openssl x509 -req -in certrequest.csr -  
signkey privatekey.pem -out certificate.pem
```

Certificate request self-signature ok

subject=C=ZA, ST=GP, L=PTA, O=APDS, OU=STUDENT, CN=NIA, emailAddress=st10396
650@vcconnect.edu.za

```
C:\Program Files\OpenSSL-Win64-2\bin>dir *.pem
```

Volume in drive C is Windows

Volume Serial Number is DCB9-6A0D

Directory of C:\Program Files\OpenSSL-Win64-2\bin

09/24/2025	01:42 PM	1,014	certificate.pem
------------	----------	-------	-----------------

09/24/2025	01:25 PM	928	privatekey.pem
------------	----------	-----	----------------

2	File(s)	1,942	bytes
---	---------	-------	-------

0	Dir(s)	94,822,023,168	bytes free
---	--------	----------------	------------

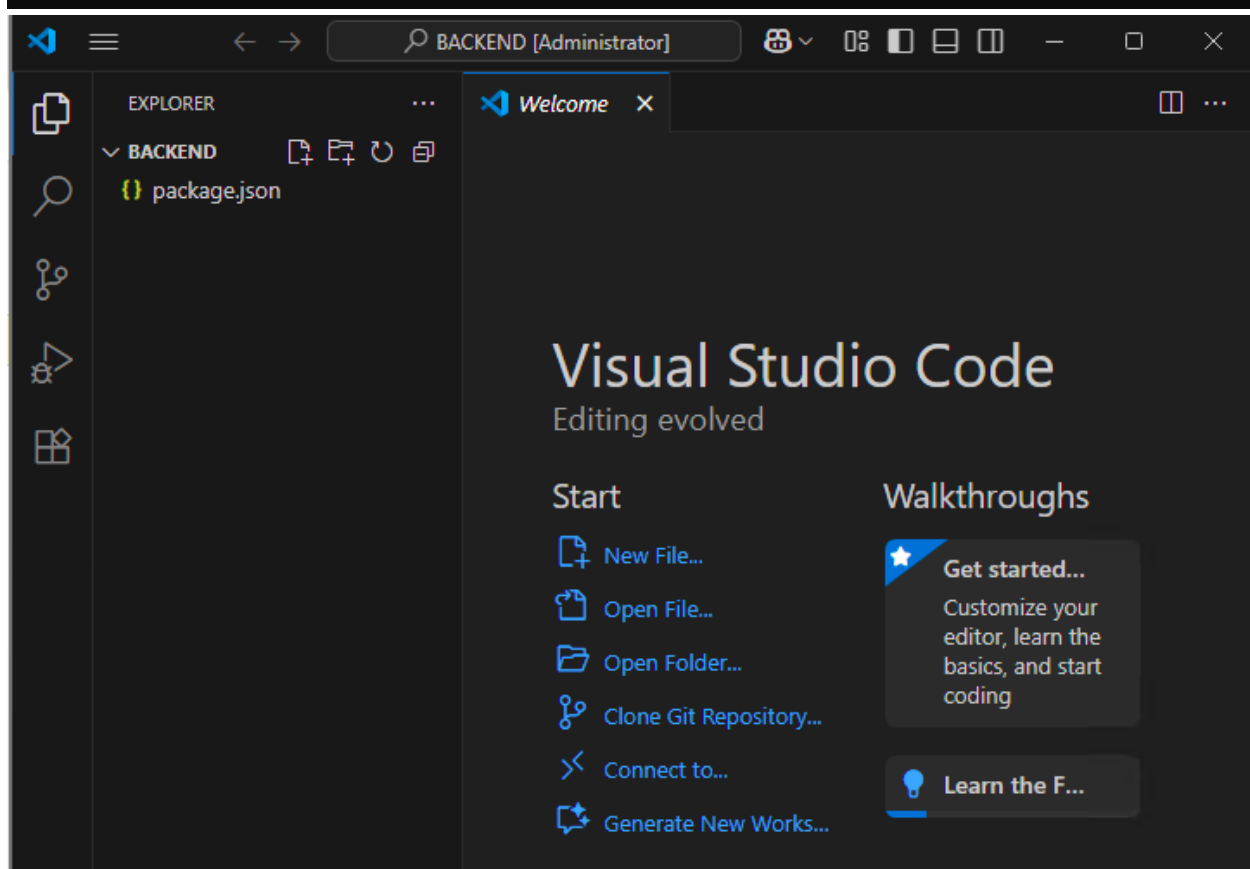
```
C:\Program Files\OpenSSL-Win64-2\bin>
```

```
Administrator: C:\Windows\Sj X + v
Microsoft Windows [Version 10.0.26100.3037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lab_services_student\Documents\BACKEND>npm init -y
Wrote to C:\Users\lab_services_student\Documents\BACKEND\package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

C:\Users\lab_services_student\Documents\BACKEND>
```



```
Administrator: C:\Windows\Sy X + v
C:\Users\lab_services_student\Documents\BACKEND>npm init -y
Wrote to C:\Users\lab_services_student\Documents\BACKEND\package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

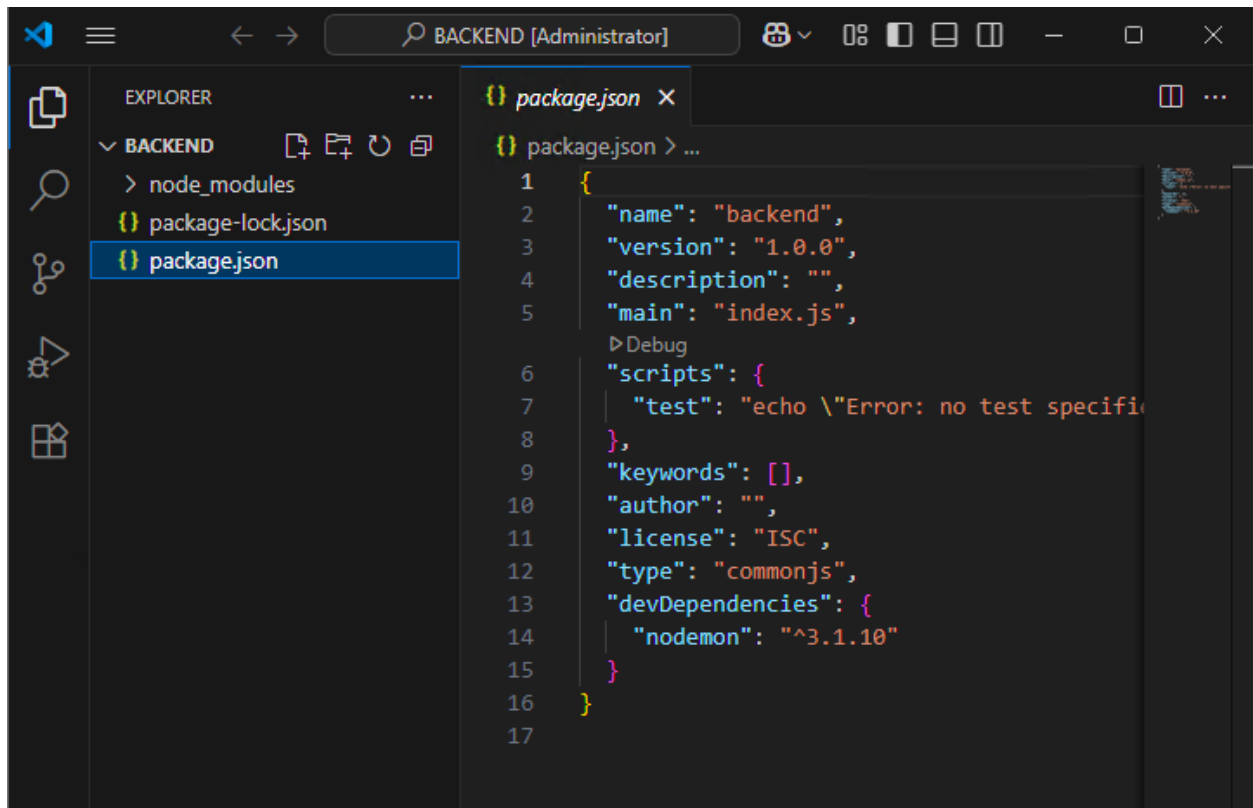
C:\Users\lab_services_student\Documents\BACKEND>npm i -D nodemon

added 30 packages, and audited 31 packages in 2s

4 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\lab_services_student\Documents\BACKEND>|
```




```
"main": "index.js",
"scripts": {
  "test": "echo \"Error: no test specified\" && exit 1"
},
"keywords": [],
"author": "",
"license": "ISC",
"type": "commonjs"
}
```

```
C:\Users\lab_services_student\Documents\BACKEND>npm i -D nodemon
```

```
added 30 packages, and audited 31 packages in 2s
```

```
4 packages are looking for funding
  run `npm fund` for details
```

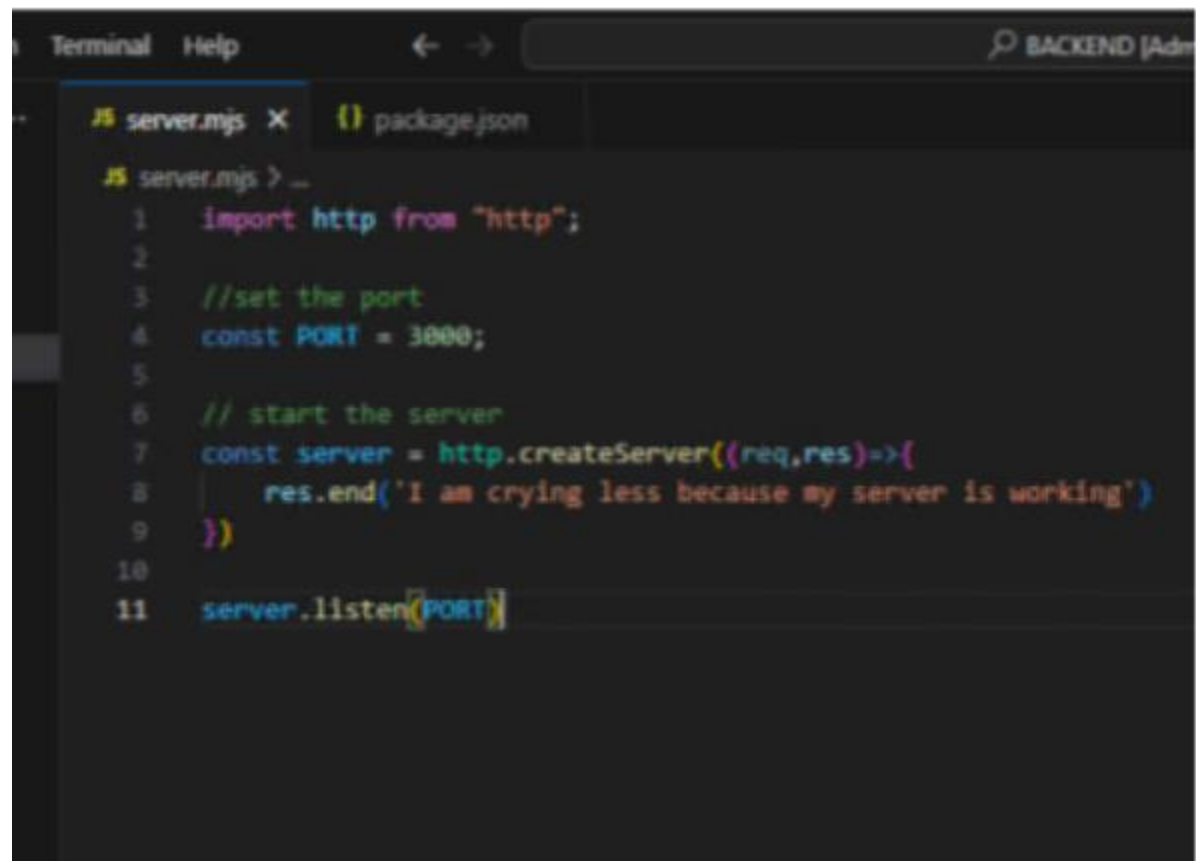
```
found 0 vulnerabilities
```

```
C:\Users\lab_services_student\Documents\BACKEND>npm run start
```

```
> backend@1.0.0 start
> node server.mjs
```

```
APDS makes me cry!
```

```
C:\Users\lab_services_student\Documents\BACKEND>|
```



The image shows a code editor window with a dark theme. At the top, there is a menu bar with 'Terminal' and 'Help' options. To the right of the menu bar is a search bar containing the text 'BACKEND [Adm...'. Below the menu bar, there are two tabs: 'server.mjs' (active) and 'package.json'. The main area of the editor displays the following JavaScript code:

```
JS server.mjs > ...  
1  import http from "http";  
2  
3  //set the port  
4  const PORT = 3000;  
5  
6  // start the server  
7  const server = http.createServer((req,res)=>{  
8    res.end('I am crying less because my server is working')  
9  })  
10  
11 server.listen(PORT)
```



npm run start



```
"author": "",  
"license": "ISC",  
"type": "commonjs"  
}
```

```
C:\Users\lab_services_student\Documents\BACKEND>npm i -D nodemon
```

```
added 30 packages, and audited 31 packages in 2s
```

```
4 packages are looking for funding  
run `npm fund` for details
```

```
found 0 vulnerabilities
```

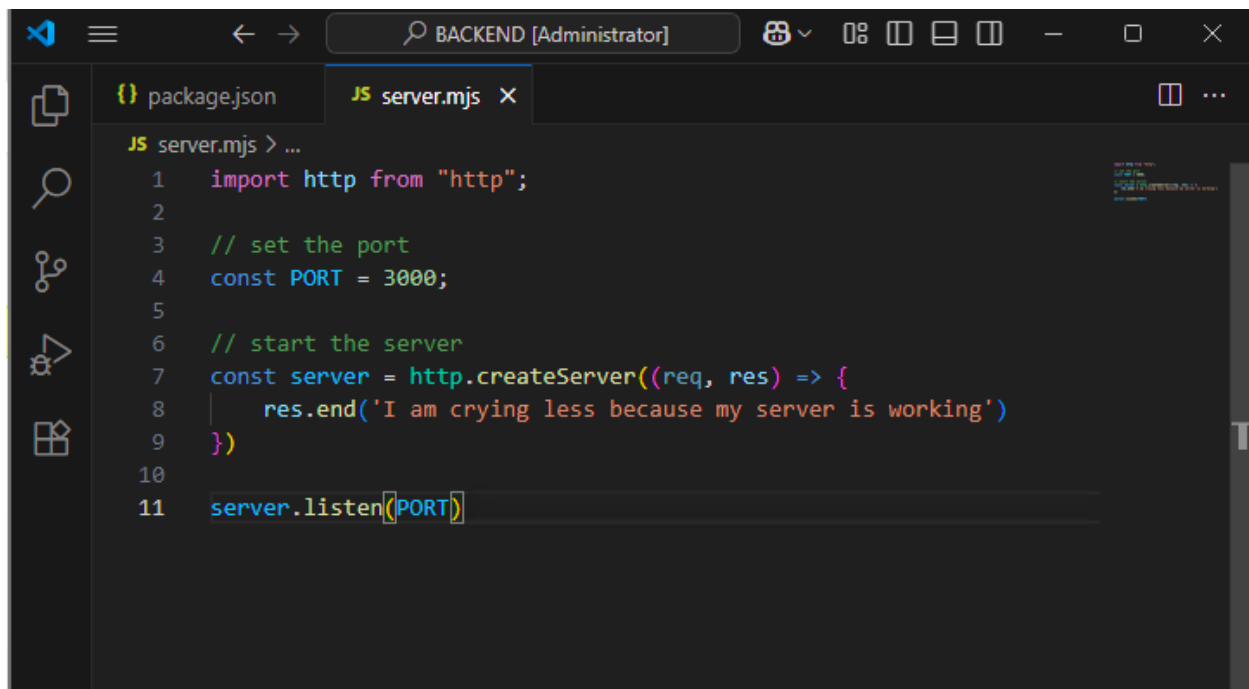
```
C:\Users\lab_services_student\Documents\BACKEND>npm run start
```

```
> backend@1.0.0 start  
> node server.mjs
```

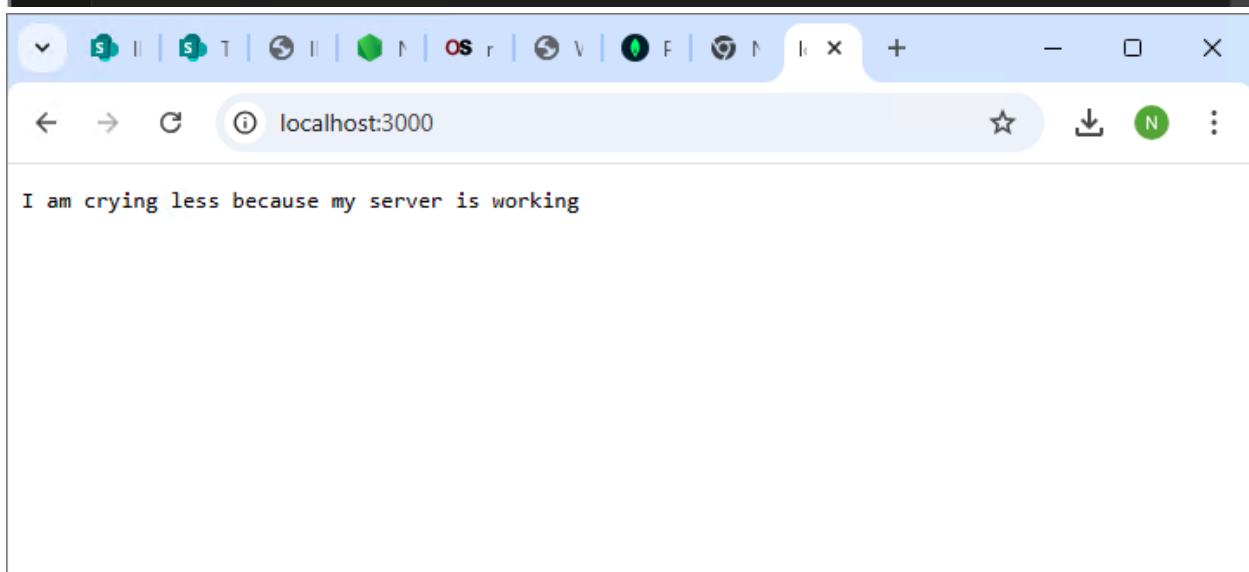
```
APDS makes me cry!
```

```
C:\Users\lab_services_student\Documents\BACKEND>npm run start
```

```
> backend@1.0.0 start  
> node server.mjs
```



```
1  import http from "http";
2
3  // set the port
4  const PORT = 3000;
5
6  // start the server
7  const server = http.createServer((req, res) => {
8    res.end('I am crying less because my server is working')
9  })
10
11  server.listen(PORT)
```



```
Administrator: C:\Windows\s  X + v
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.mjs`
node:events:486
  throw er; // Unhandled 'error' event
  ^

Error: listen EADDRINUSE: address already in use :::3000
    at Server.setupListenHandle [as _listen2] (node:net:1940:16)
    at listenInCluster (node:net:1997:12)
    at Server.listen (node:net:2102:7)
    at file:///C:/Users/lab_services_student/Documents/BACKEND/server.mjs:11:8
    at ModuleJob.run (node:internal/modules/esm/module_job:371:25)
    at process.processTicksAndRejections (node:internal/process/task_queues:105:5)
    at async onImport.tracePromise.__proto__ (node:internal/modules/esm/loader:669:26)
    at async asyncRunEntryPointWithESMLoader (node:internal/modules/run_main:101:5)
Emitted 'error' event on Server instance at:
    at emitErrorNT (node:net:1976:8)
    at process.processTicksAndRejections (node:internal/process/task_queues:90:21) {
  code: 'EADDRINUSE',
  errno: -4091,
  syscall: 'listen',
  address: ':::',
  port: 3000
}

Node.js v24.6.0
[nodemon] app crashed - waiting for file changes before starting...
```

 package.json

 server.mjs

 Extension: Postman 

 ...



Postman

Postman  postman.com |  1,937,964 |   

Streamline API development and testing with the po...

Installing

☒ Auto Update 

DETAILS

FEATURES

CHANGELOG

The Postman VS Code extension

With the Postman VS Code extension, you can develop and test your APIs in Postman right from [Visual Studio Code](#) and other supported applications. You can use the VS Code extension to send API requests, manage collection and environments, troubleshoot requests, and much more. Streamline your development workflow by testing your APIs in the same application you use to develop them.

The VS Code extension also supports [Visual Studio Code Insiders](#), [VSCodium](#), [Cursor](#), and [Windsurf](#).

The VS Code extension supports version 1.74 and later of the Visual Studio Code, Visual Studio Code Insiders, and VSCodium applications. The VS Code extension supports all versions of Cursor and Windsurf.

To learn more about the Postman VS Code extension and its capabilities, [check out the documentatic](#) 

Prerequisites

Marketplace

Identifier	postman.postman-for-vscode
Version	1.16.2
Published	2023-05-11, 18:39:05
Last Released	2025-09-18, 09:55:22

Categories

Programming Languages

Education

Testing

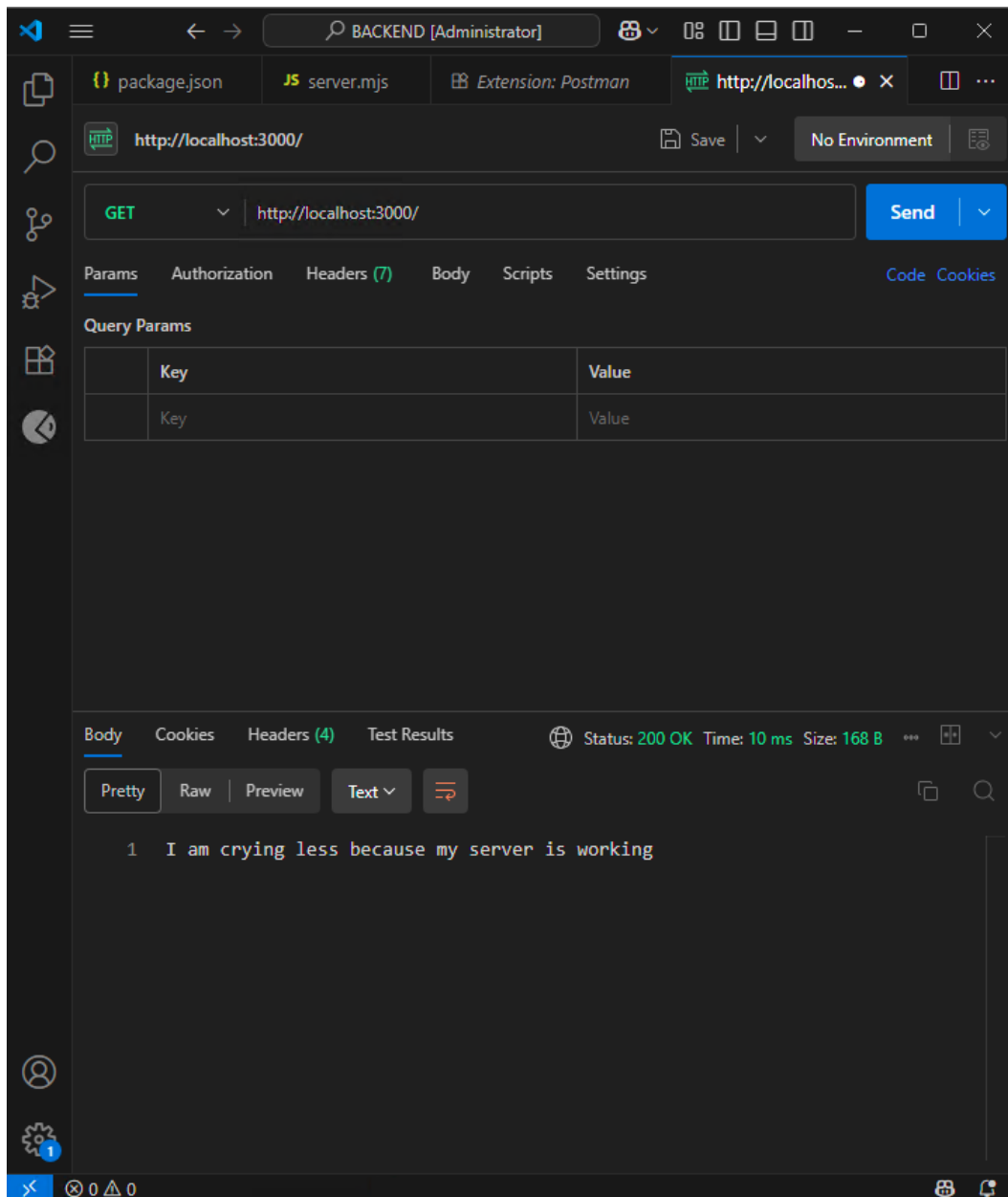
Resources

[Marketplace](#)
[Issues](#)
[Repository](#)
[Postman](#)

   0



```
Administrator: C:\Windows\Sy X + v
Microsoft Windows [Version 10.0.26100.3037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lab_services_student\Documents\BACKEND>npm i express

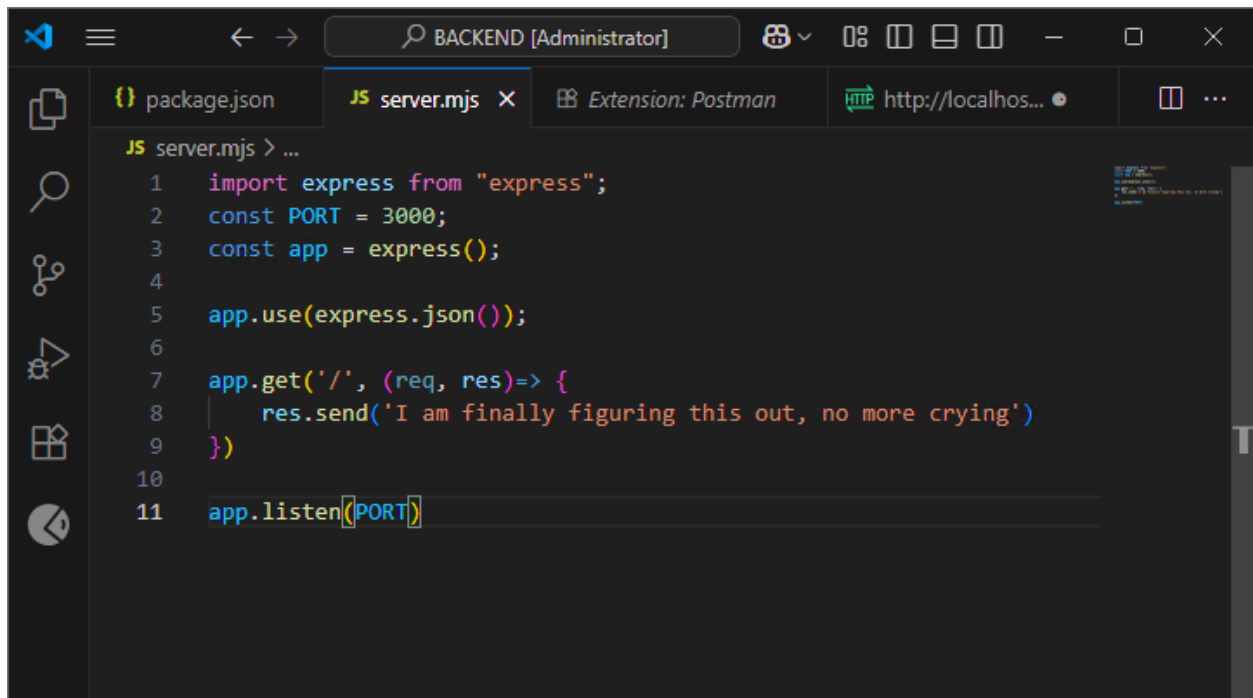
added 66 packages, and audited 97 packages in 3s

20 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

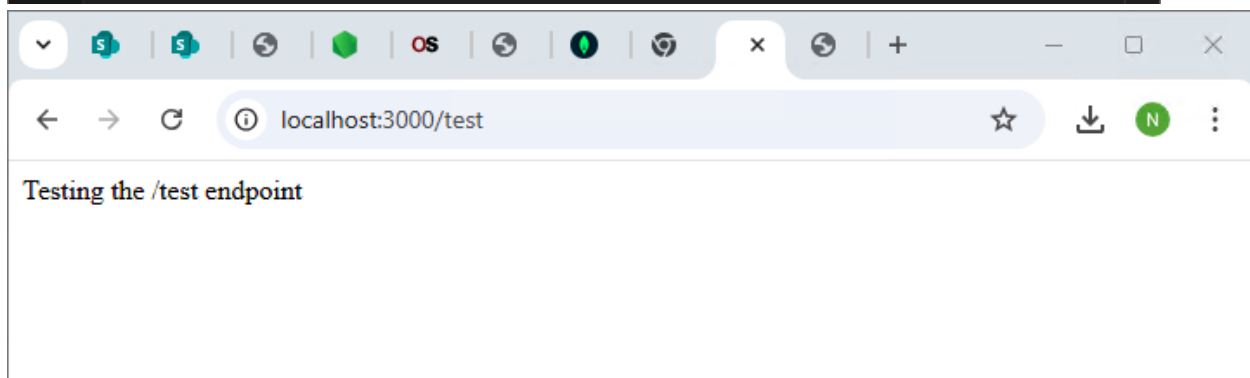
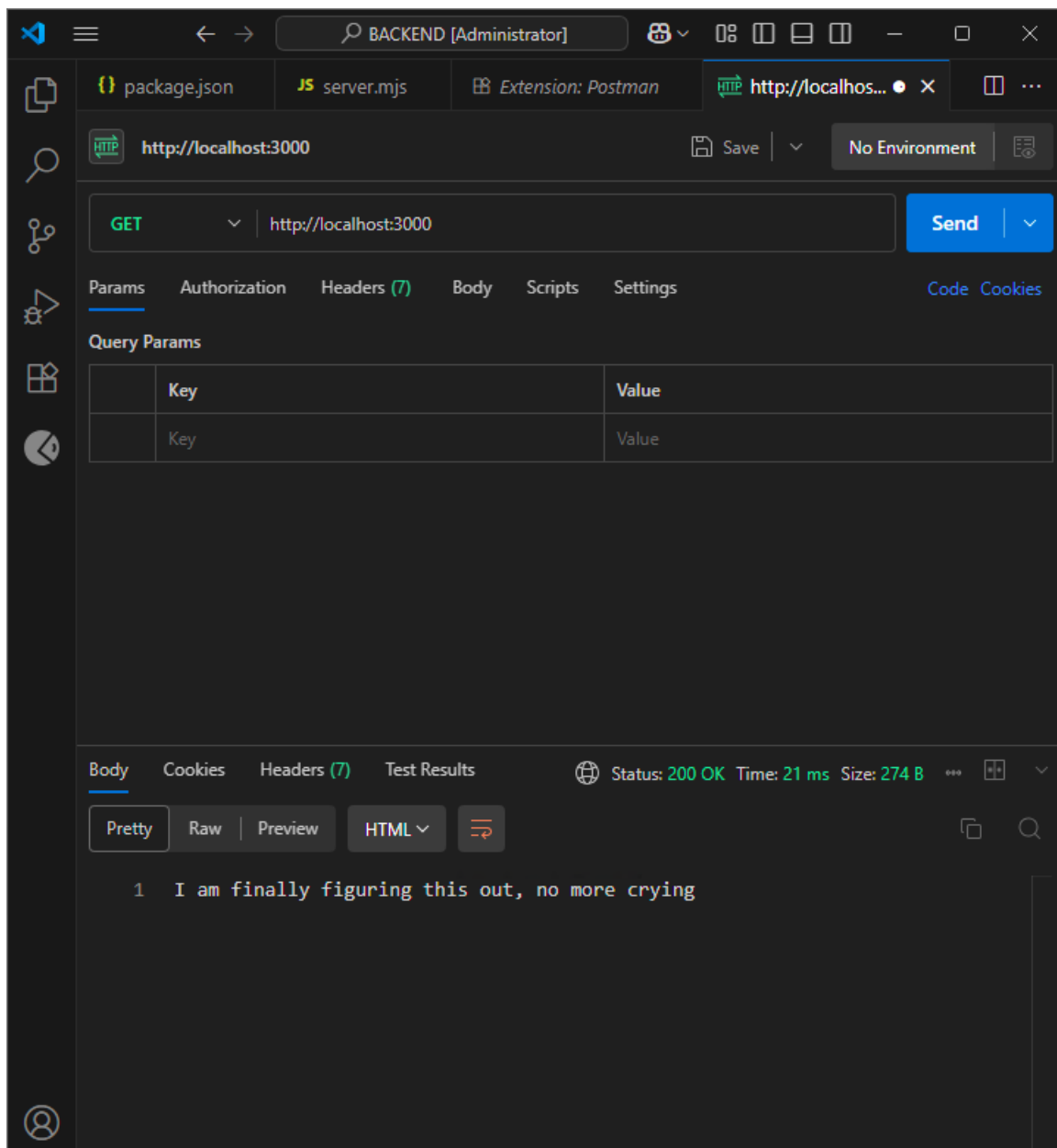
C:\Users\lab_services_student\Documents\BACKEND>
```

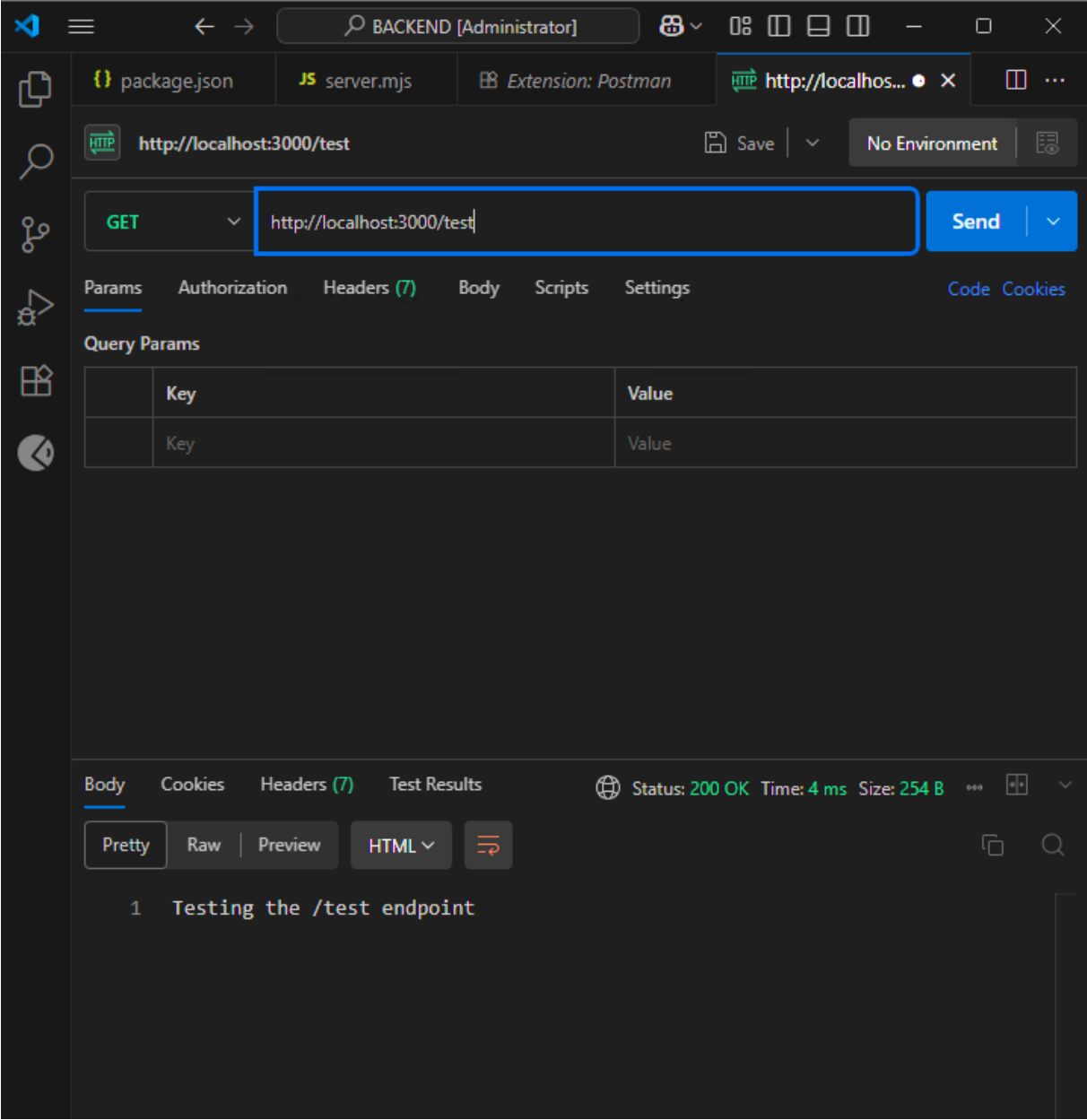
```
VS Code BACKEND [Administrator]
package.json JS server.mjs x Extension: Postman http://localhost...
JS server.mjs > ...
1 import express from "express";
2 const PORT = 3000;
3 const app = express();
4
5 app.use(express.json());
6
7 app.get('/', (req, res)=> {
8   res.send('I am finally figuring this out, no more crying')
9 })
10
11 server.listen(PORT)
```

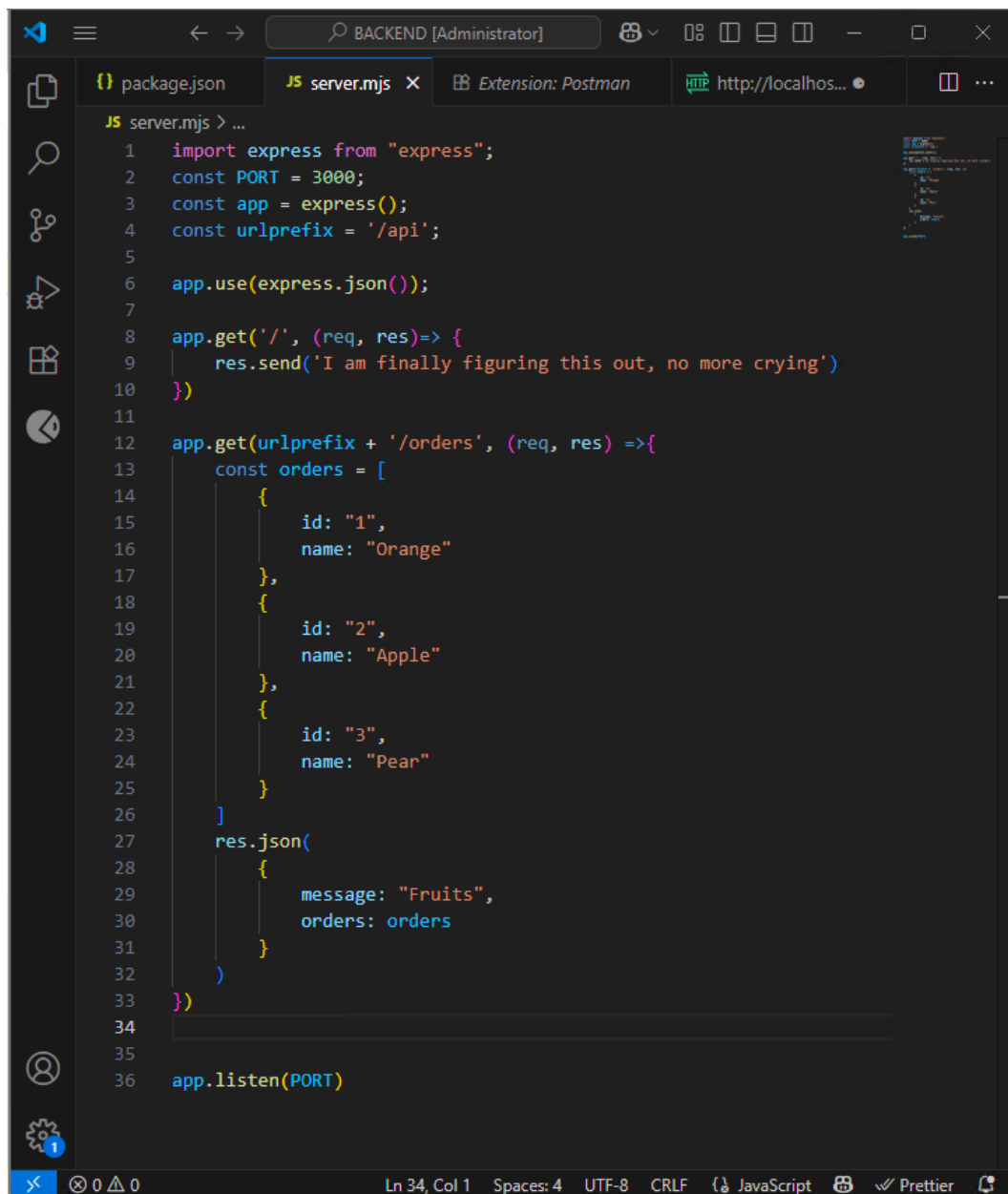



The image shows a Visual Studio Code editor window with a dark theme. The title bar at the top reads "BACKEND [Administrator]". The Explorer sidebar on the left shows a file named "package.json" and a folder named "server.mjs". The main editor area displays the contents of "server.mjs", which is a JavaScript file for creating a simple web server using Express.js. The code is as follows:

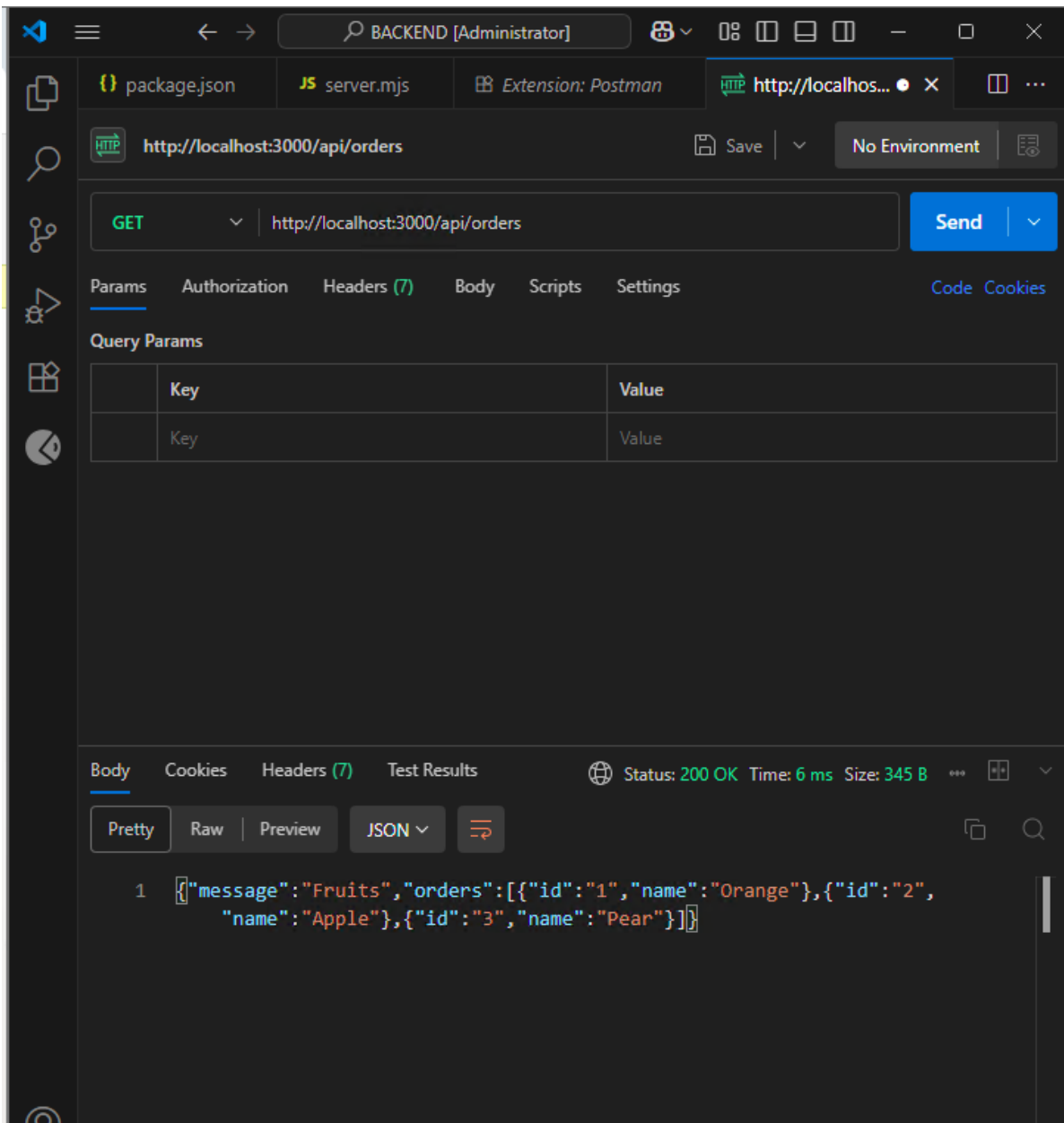
```
JS server.mjs > ...
1  import express from "express";
2  const PORT = 3000;
3  const app = express();
4
5  app.use(express.json());
6
7  app.get('/', (req, res)=> {
8    res.send('I am finally figuring this out, no more crying')
9  })
10
11 app.listen(PORT)
```

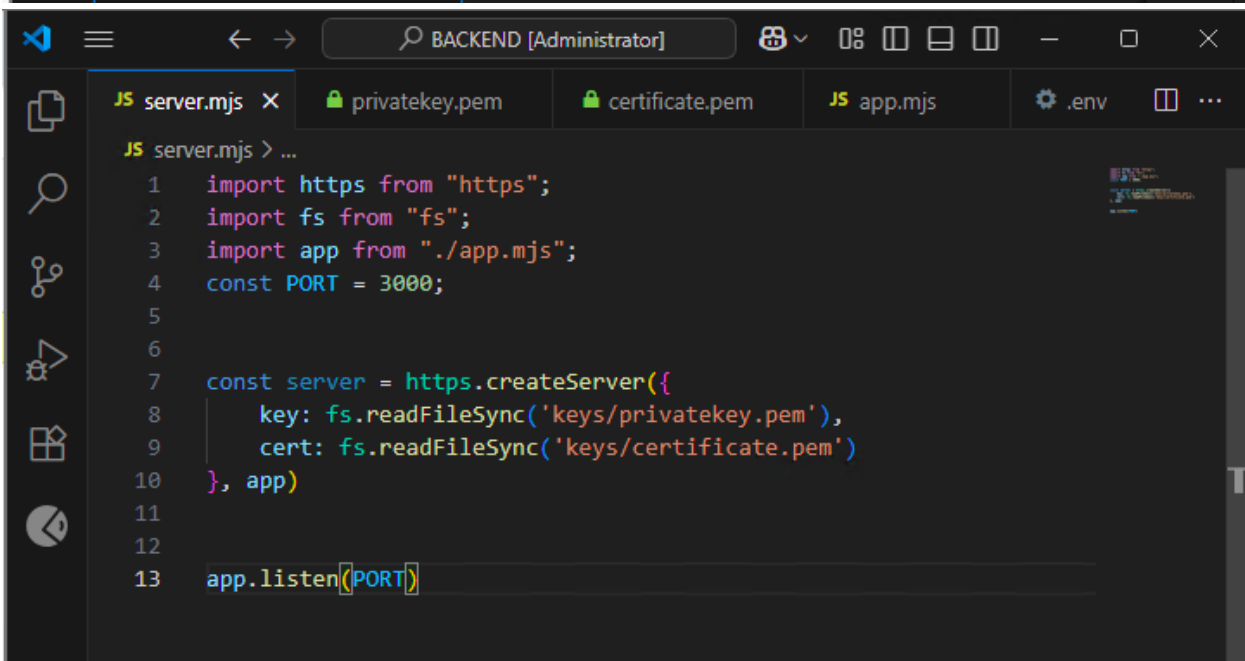
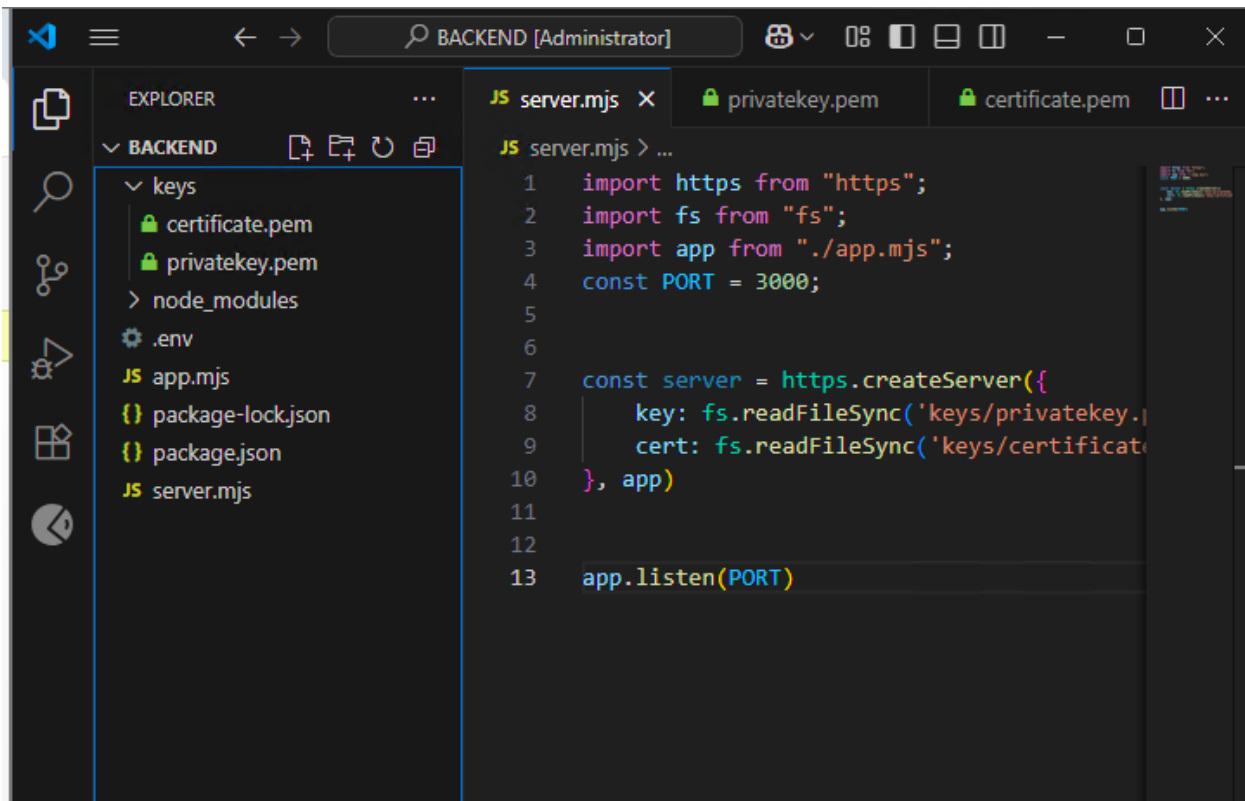






```
1  import express from "express";
2  const PORT = 3000;
3  const app = express();
4  const urlprefix = '/api';
5
6  app.use(express.json());
7
8  app.get('/', (req, res)=> {
9    res.send('I am finally figuring this out, no more crying')
10 })
11
12 app.get(urlprefix + '/orders', (req, res) =>{
13   const orders = [
14     {
15       id: "1",
16       name: "Orange"
17     },
18     {
19       id: "2",
20       name: "Apple"
21     },
22     {
23       id: "3",
24       name: "Pear"
25     }
26   ]
27   res.json(
28     {
29       message: "Fruits",
30       orders: orders
31     }
32   )
33 })
34
35
36 app.listen(PORT)
```





Administrator: C:\Windows\Sy X + v

Microsoft Windows [Version 10.0.26100.3037]
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C:\Users\lab_services_student\Documents\BACKEND>npm i mongodb mongoose dotenv

added 14 packages, and audited 111 packages in 6s

21 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

C:\Users\lab_services_student\Documents\BACKEND>